A Type 2 Diabetes Digital Twin

Fusing synthetic EHR + wearable CGM data to predict glucose spikes 2 hours ahead

Digital Twin Challenge 2026 — Happiest Health

Problem Statement

- India has ~101 million people with diabetes and ~136 million with prediabetes (ICMR-INDIAB, 2023)

- Care is reactive: patients are seen every 3 months, HbA1c is a lagging indicator

- Post-meal glucose spikes drive long-term complications (heart, kidney, eye, nerve damage)

- CGMs and smartwatches produce data, but doctors have no tool that fuses it with the EHR

  and warns them before an adverse event happens

Healthcare Use Case

- Condition: Type 2 Diabetes (one of India's most common chronic conditions)

- Adverse event: hyperglycaemic spike (CGM > 180 mg/dL) predicted 2 hours in advance

- Doctor: sees each virtual patient's EHR + live CGM / HR / HRV / steps / sleep in one view

- Early warning -> nudge a 15-min walk, defer carbs, review the pre-meal dose

- What-if simulation on the twin: test an intervention before advising the patient

- Population view: rank patients by risk for remote monitoring in under-served districts

Data Streams (Sandbox Rules)

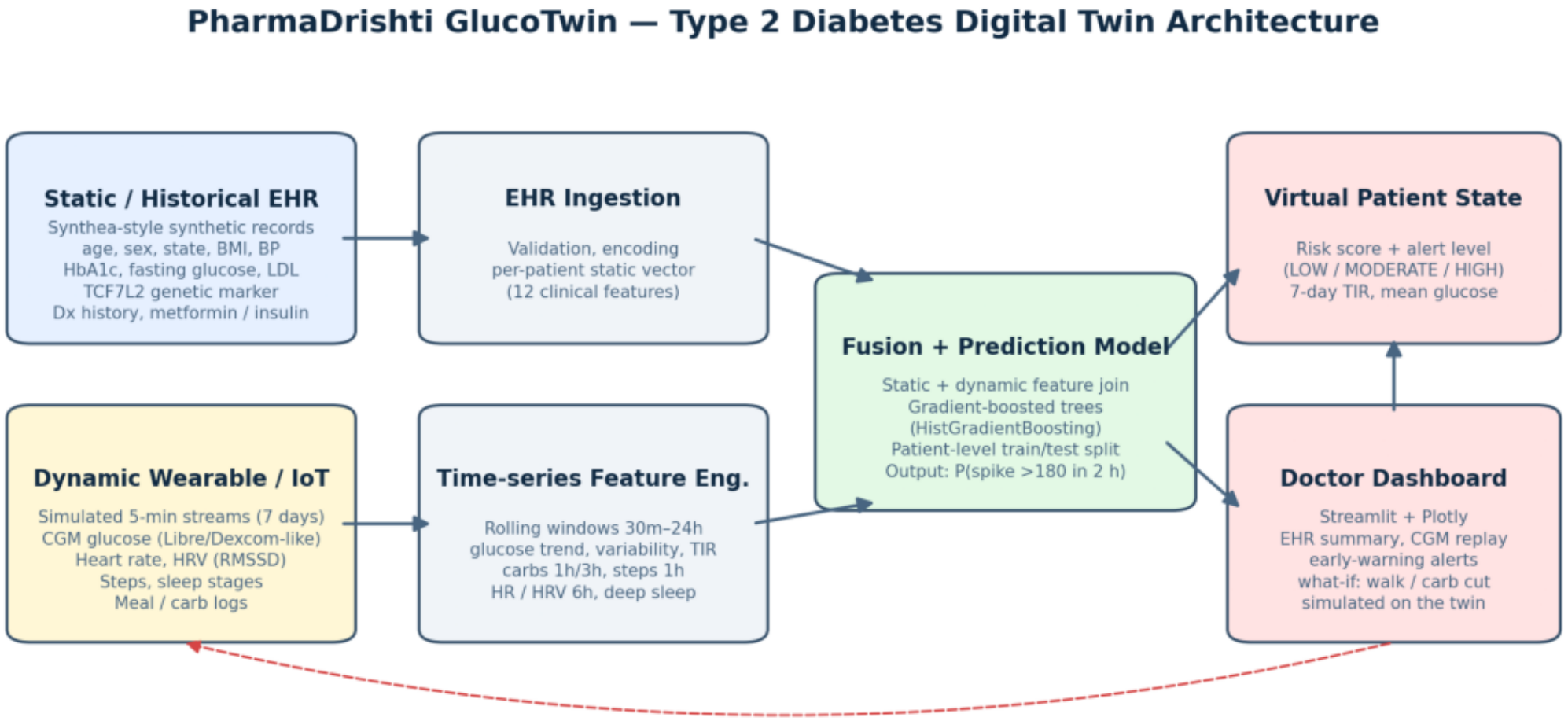
1. Static EHR (synthetic, Synthea-style): demographics, state, BMI, BP, HbA1c, fasting glucose, LDL, TCF7L2 risk alleles, family history, T2D / hypertension Dx, metformin / insulin

2. Dynamic wearables (simulated, 5-min): CGM glucose, heart rate, HRV (RMSSD), steps, sleep stages, meal carb logs. Model includes meal absorption, insulin resistance, activity effect, dawn phenomenon, poor-sleep stress

- 200 virtual patients x 7 days = 332,162 labelled 5-min windows

- No real patient data -> compliant with DPDP Act / HIPAA

Architecture



Held-out patients: ROC-AUC 0.96 · recall 0.876 · PR-AUC 0.803 | Privacy: 100% synthetic data (DPDP / HIPAA safe)

AI/ML Model

- Fusion: 12 static EHR features joined with 14 rolling wearable features at each time step

- Model: gradient-boosted decision trees (scikit-learn HistGradientBoostingClassifier),

  class-balanced, 300 iterations

- Label: glucose currently <= 180 and > 180 at any point in the next 120 minutes

- Validation: patient-level split - 20% of virtual patients never seen in training

- Ablation shows the value of fusing both data streams (next slide)

Outcomes & Results

- Fused model ROC-AUC: 0.96 | PR-AUC: 0.803

- Recall 0.876 · Precision 0.586 · F1 0.702 (threshold 0.5, spike rate 13.6%)

- Ablation: wearable-only AUC 0.957, EHR-only AUC 0.935 -> fusion is best

- Most spikes are flagged up to 2 h ahead -> time to act

- Working doctor dashboard: CGM replay, alerts, what-if intervention simulator

- Trains in ~15 s on a laptop; inference in milliseconds

Tech Stack

- Python 3.10+, NumPy, Pandas

- scikit-learn (HistGradientBoostingClassifier), joblib

- Streamlit + Plotly doctor dashboard

- Matplotlib (architecture diagram & slides)

- Existing PharmaDrishti modules: XGBoost, SHAP, NetworkX knowledge graph of

  NPPA + FDA Orange Book data for therapy affordability & access analysis

Limitations & Roadmap

- Synthetic physiology - next: validate on OhioT1DM / open CGM datasets and MIMIC-IV EHRs

- Add hypoglycaemia (<70) and cardiovascular-anomaly (HRV + sleep) predictions

- Personalised fine-tuning per patient as more twin data arrives

- FHIR / ABDM (Ayushman Bharat Digital Mission) integration for real EHR ingestion

- Link to PharmaDrishti pricing engine: recommend affordable therapy per patient segment